Assignment 4

1a) $P[X \geq a] \leq \frac{E[X]}{a}$

Quicksort takes $O(n \log n)$ expected work so

$E[X] = O(n \log n)$

Probability that Quicksort does $\Omega(n^2)$?

$P[X \geq \Omega(n^2)]$

$\Rightarrow P[X \geq c_1 n^2] \leq \frac{c_2 n \log n}{c_1 n^2}$.

following the formula, adding constants because big-O and omega "hides" them or takes all $n \log n$ and n^2 with any constant.

$\Rightarrow \leq \frac{c_2}{c_1} \cdot \frac{\log n}{n}$ back to big O: $\therefore O(\frac{\log n}{n})$

1b $P[X \geq a] \leq \frac{E[X]}{a}$

$P[X \geq 10^c n \lg n] \leq \frac{c_2 n \log n}{10^c n \log n} = \frac{c_2}{10^c}$

so $O(\frac{1}{10^c})$ or $O(1)$

Even though $n \log n$ is present in the treshold, it cancels out. The probability of having more work than the expected work drops experientially based on the scaling factor c

- The probability depends only on c, and not on the input size n (b/c it cancels out). this means no matter how many items we sort, the risk of the algorith running 10^c times slower is small and bounded by this same small number

2a

Description

We want su

- we can run

success vs

1 Find

to get a

$(1-\delta$ is

2 loop

- run a

- check

- if C

- if its all

For A' to

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work is

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worst

2a

Description of A'

We want success probability of δ

- we can run independent trials of A and verify its success vs failure:

1. find minimum number of trials k it takes to get a failing probability of less than $1-\delta$ ($1-\delta$ is prob. of failing, δ prob of success).
2. loop from $i = 1$ to k
   - run algorithm and get the output $A(I)$.
   - check if its success or failure $C(A(I))$
   - if $C(s)$ is success → stop and return $A(I)$.
- if its all failure, return a fail signal.

for A' to fail all k trials:

probability of that is $(1-\epsilon)^k$

we want the failure prob to be less than $1-\delta$ (or equal).

$$(1-\epsilon)^k \leq 1-\delta$$

$$\ln \qquad \ln$$

$$k \ln(1-\epsilon) \leq \ln(1-\delta)$$

negative

$$k \geq \frac{\ln(1-\delta)}{\ln(1-\epsilon)}$$

flip↑

thats how many k trials to get the prob we want

Work of A'

work per iteration = $O(w(n)) + O(v(n))$

since $v(n) \in O(w(n))$, each iteration takes $O(w(n))$ work. Multiplying times k iterations.

$$= O\left(w(n) \cdot \frac{\ln(1-\delta)}{\ln(1-\epsilon)}\right).$$

2b.

We can make the loop infinite until a correct solution is verified

1. start infinite loop
2. Execute trial:
   - run the algorithm A(I)
   - check C(S)
3. if C(S) is a success return A(I) and end the algorithm
4. if C(S) is a failure, go to next iteration

Let number of trials be k

each run of A succeeds independently with prob. ϵ.

number of n trials until first success is a geometric distribution

by def

$$E[k] = \frac{1}{\epsilon}$$

from part a

$$O(W(n)) + O(V(n)) = O(W(n))$$

so work · trials:

expected work: $E[k] \cdot$ Work per trial $= \frac{1}{\epsilon} \cdot O(W(n))$

$$= O\left(\frac{W(n)}{\epsilon}\right)$$